

GNSS Signal Authentication via Power and Distortion Monitoring

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Abstract—We propose a simple low-cost technique that enables civil Global Positioning System (GPS) receivers and other civil global navigation satellite system (GNSS) receivers to reliably detect carry-off spoofing and jamming. The technique, which we call the Power-Distortion detector, classifies received signals as interference-free, multipath-afflicted, spoofed, or jammed according to observations of received power and correlation function distortion. It does not depend on external hardware or a network connection and can be readily implemented on many receivers via a firmware update. Crucially, the detector can with high probability distinguish low-power spoofing from ordinary multipath. In testing against over 25 high-quality empirical data sets yielding over 900,000 separate detection tests, the detector correctly alarms on all malicious spoofing or jamming attacks while maintaining a $<0.5\%$ single-channel false alarm rate.

Index Terms—satellite navigation systems, Global Positioning System, Global Navigation Satellite Systems, navigation security, GNSS spoofing, GNSS jamming

I. INTRODUCTION

GNSS receivers are tremendously popular in navigation and timing applications due to their accuracy, low cost, and global operation. Low cost can be credited to the fact that civil GNSS signals are defined in freely-available, open-access standards [1], [2], which makes receiver development straightforward. But an open-access standard, together with civil GNSS signals' near-perfect predictability, invites forgery: receivers can fall victim to spoofing attacks in which counterfeit GNSS signals fool the receiver into reporting a hazardously misleading position or time [3]. The vulnerability of civil GNSS receivers to spoofing is a serious risk for GNSS-dependent critical infrastructure and safety-of-life applications [4], [5].

GNSS authentication techniques can be broadly categorized into three groups: (1) cryptographic techniques that exploit unpredictable but verifiable signal modulation in the GNSS

spreading code or navigation data, (2) geometric techniques that exploit the angle-of-arrival diversity of authentic GNSS signals, and (3) GNSS signal processing techniques that do not fall into categories (1) or (2). A comprehensive survey of GNSS authentication techniques is offered in [6].

Cryptographic signal authentication is effective [6]–[8], but, despite recent interest and engagement by U.S. and European satellite navigation agencies [9], [10], no open civil GNSS signals yet incorporate cryptographic modulation. Moreover, it has become clear that financial and technical hurdles will impede development and implementation of such modulation for years to come. It is possible to leverage the existing encryption of military GNSS signals for civil signal authentication [11], but this technique requires the protected receiver to be connected to a data network, an undesirable dependency that prevents stand-alone operation.

Authentication techniques that exploit GNSS signals' geometric diversity can also be highly effective. These include angle-of-arrival discrimination techniques based on multiple antennas [12]–[15], or a single antenna experiencing oscillatory motion [16], or a single antenna with multiple feeds [17]. The drawback of these approaches is their reliance on multiple antennas, antenna motion, or an assumption that interference signals arrive from below the antenna's horizon. Likewise, methods that require coupling with inertial sensors [18], [19] or vision sensors may prove impractical in applications with cost, size, weight, or power constraints.

Practical near-term GNSS signal authentication techniques are those that do not require changes to GNSS signals-in-space, are receiver-autonomous, low-cost, require no additional hardware, and can be implemented via a software or firmware update. Recognizing the value of techniques that fall into this category, previous work has proposed monitoring the total received power via the Automatic Gain Control (AGC) setpoint [20], and monitoring autocorrelation profile distortion [21]–[23]. But when acting separately these techniques are unreliable for signal authentication. A received power monitor that ignores correlation distortion may not detect a low-power spoofer. Moreover, because a power-monitoring-only technique does not distinguish between spoofing and jamming, its alarm rate can become intolerable in urban areas where so-called personal privacy devices (PPDs, small GNSS jammers) [24] are common. For their part, the distortion-monitoring approaches in [21]–[23] ignore total received power, and thus can be fooled by a spoofer transmitting with a significant power advantage over the authentic signals, which, by action of the AGC, forces the authentic signals under the

Manuscript submitted on DD MMM 2017

K. Wesson was supported in part by DoD, Air Force Office of Scientific Research, National Defense Science and Engineering Graduate Fellowship 32 CFR §168a, and The University of Texas at Austin Microelectronics and Computer Development Fellowship. J. Gross was supported in part by a West Virginia University Big XII Faculty Fellowship. T. Humphreys was supported in part by the National Science Foundation under Grant No. 1454474 (CAREER).

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noise floor, leaving a distortion-free correlation function [8].

A GNSS authentication technique is needed that is both practical in the sense described above and reliable at detecting spoofing. To this end, we propose to combine the elemental tests mentioned earlier, namely, detection of anomalous received power and detection of correlation profile distortion, in a GNSS signal authentication technique we call the Power-Distortion detector, or PD detector for short. The key insight behind our approach is this: The practically-unavoidable interaction between authentic and spoofed GNSS signals during the initial stages of a tracking-points-carry-off spoofing attack makes such spoofing evident, with high probability, in received power or signal distortion or both. If a carry-off-type spoofer transmits at a low signal power, the attack will either be ineffective or will cause significant correlation function distortion as the similarly-sized spoofing and authentic signals interact. On the other hand, if a spoofer transmits at a high signal power, the correlation function may be distortion-free but the receiver's total received power will be anomalously high. Our proposed technique traps a would-be spoofer between simultaneous measurements of received power and correlation function distortion. Provided the spoofer is unable to block or otherwise null the authentic GNSS signals impinging on the receiver's antenna (a difficult task if the receiver enjoys a physical security perimeter [6]), the combination of these measurements within a Bayesian detection framework poses a formidable defense against carry-off-type spoofing.

This paper, a significant extension of our work in [25], makes three main contributions. First, it introduces a novel technique for detecting GNSS jamming and carry-off-type spoofing and rigorously develops the measurement models and probability distributions required to characterize the detection statistic. Second, it presents a Monte-Carlo-type method for determining the Bayes-optimal decision rule and offers detailed consideration of the requisite cost function. Third, it presents a thorough evaluation of the proposed technique against three realistic data sets: (1) the Texas Spoofing Test Battery [26]–[28], a public set of GPS spoofing recordings; (2) the RNL Multipath and Interference Recordings [29], [30], a public set of deep urban GNSS recordings with significant multipath; and (3) a set of recordings of GNSS signals subject to jamming.

II. SIGNAL MODELS

A. Pre-Correlation Model

Consider the following generic representation of an authentic GNSS signal exiting a receiver's radio frequency (RF) front-end downconversion chain. For notational compactness, the signal is expressed by its complex baseband representation,

$$r_A(t) = \sqrt{P_A} D(t - \tau_A) C_r(t - \tau_A) \exp(j\theta_A) \quad (1)$$

where t is time in seconds, P_A is the received power of the authentic signal in Watts, $D(t)$ is the ± 1 -valued navigation data modulation, $C_r(t)$ is the ± 1 -valued pseudorandom spreading (ranging) code, τ_A is the code phase in seconds, and $\exp(j\theta_A)$ is the carrier with phase θ_A in radians. P_A , τ_A , and θ_A are assumed to vary with time; their time dependency is

suppressed for notational simplicity. Without loss of generality for the purposes of this paper, the navigation data modulation $D(t)$ can be ignored; hence, hereafter we assume $D(t) = 1$.

Let $r_I(t)$ represent a single complex-valued interference signal that is structurally identical to $r_A(t)$. This could be a multipath, spoofing, or jamming signal. If multipath, $r_I(t)$ represents the strongest reflection at time t , whose effect on received power and correlation function distortion is a good proxy for that of the aggregate multipath. If jamming, $r_I(t)$ represents structured-signal jamming, an especially potent form of jamming similar to spoofing except there is no expected correlation between the jamming signal's code or carrier phase and those of $r_A(t)$ [31]. The interference signal is modeled as

$$r_I(t) = \sqrt{\eta P_A} C_r(t - \tau_I) \exp(j\theta_I) \quad (2)$$

where $\eta = P_I/P_A$ is the interference signal's power advantage relative to the authentic signal, and τ_I and θ_I are the interference signal's code and carrier phase, respectively.

To complete the received signal model, let

$$r_N(t) = N(t) + M(t) \quad (3)$$

be a white zero-mean complex-valued Gaussian process that models the sum of thermal noise $N(t)$, with constant spectral density N_0 , and multi-access interference $M(t)$, with variable spectral density M_0 . The two noise components are assumed to be independent so that $r_N(t)$ has density $N_0 + M_0$. $M(t)$ accounts for the noise contribution from other legitimate GNSS signals besides the desired signal (collectively called multi-access signals), and any interference accompanying these. With the addition of $r_N(t)$, the full received signal-plus-interference-and-noise model is given by

$$r(t) = r_A(t) + r_I(t) + r_N(t) \quad (4)$$

As shown in Fig. 1, an AGC circuit is assumed to apply a scaling factor $\beta(t)$ to $r(t)$ so that the power in the scaled signal $\beta(t)r(t)$ remains constant. Subsequent to AGC scaling, the signal is quantized and encoded; for simplicity, these operations are ignored in Fig. 1 and in the remainder of this paper, as their effects on the this paper's detection processing are negligible.

At the core of GNSS signal processing is correlation of $\beta(t)r(t)$ with a local replica

$$\ell(t, \tau) = C_\ell(t - \hat{\tau} - \tau) \exp(j\hat{\theta})$$

where τ is an arbitrary code phase lag in seconds and $C_\ell(t)$ is the local code replica, which, ignoring the effects of band-limiting on the received signal, is often made equal to $C_r(t)$. The goal of a receiver's code and carrier tracking loops is to drive the estimates $\hat{\tau}$ and $\hat{\theta}$ to match τ_A and θ_A as accurately as possible. In practice, however, $\hat{\tau}$ and $\hat{\theta}$ track the code and carrier phase of the composite signal $r(t)$, not just those of $r_A(t)$.

B. Post-Correlation Model

Correlation and accumulation over an interval T ending at time $t_k = kT$, $k \in \{1, 2, \dots\}$ produce the complex-valued

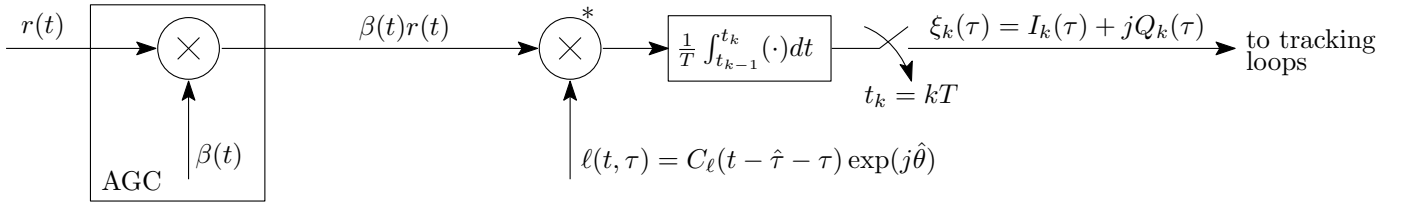


Fig. 1: Block diagram of the standard AGC, correlation, and accumulation operations in a GNSS receiver. The product of the AGC-scaled incoming signal $\beta(t)r(t)$ and the conjugate of the local replica $\ell(t, \tau)$ is accumulated over T seconds to produce the discrete complex-valued accumulation product $\xi_k(\tau)$. For notational convenience, the accumulation product has been scaled by $1/T$.

accumulation product ξ_k , which, when viewed as a function of the arbitrary lag τ introduced by the local replica $\ell(t, \tau)$, is called the receiver's correlation function for signal $r_A(t)$ at time t_k , and is modeled as [32]

$$\xi_k(\tau) = \beta_k[\xi_{Ak}(\tau) + \xi_{Ik}(\tau) + \xi_{Nk}(\tau)] \quad (5)$$

where β_k is the average value of $\beta(t)$ over the k th accumulation interval, and $\xi_{Ak}(\tau)$, $\xi_{Ik}(\tau)$, $\xi_{Nk}(\tau)$ are the complex correlation function components corresponding to the authentic signal, the interference signal, and thermal noise, respectively. Fig. 2 illustrates such a correlation function.

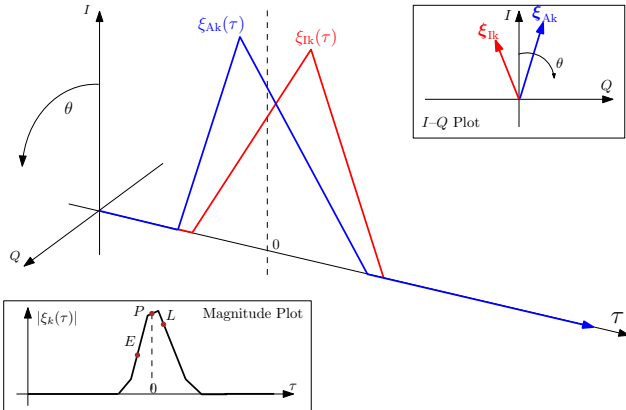


Fig. 2: Components of the correlation function $\xi_k(\tau)$ for an example scenario with a triangular-shaped $R(\tau)$ and a strong spoofing or multipath interference component $\xi_{Ik}(\tau)$. For visual clarity, the thermal noise component $\xi_{Nk}(\tau)$ is not shown. The upper-right inset shows the phase angle of the authentic and interference components relative to the in-phase I and quadrature Q components of the local replica $\ell(t, \tau)$. The lower-left inset shows the magnitude of the composite correlation function $\xi_k(\tau)$, which is clearly distorted by the interference component. E , P , and L mark the early, prompt, and late correlation taps, respectively. The local replica's code phase estimate $\hat{\tau}$ attempts to track the code phase of the composite signal $r(t)$ by equalizing E and L . The prompt tap P is located at $\tau = 0$.

The function $R(\tau) = E[C_r(t)C_\ell(t - \tau)]$, often called the autocorrelation function of $C_r(t)$ even though, strictly speaking, $C_\ell(t)$ may be slightly different from $C_r(t)$, approximates the interaction between $C_r(t)$ and $C_\ell(t)$ over the correlation

and accumulation operations:

$$R(\tau) \approx \frac{1}{T} \int_{t_{k-1}}^{t_k} C_r(t)C_\ell(t - \tau)dt$$

The correlation components $\xi_{Ak}(\tau)$ and $\xi_{Ik}(\tau)$ can be modeled in terms of $R(\tau)$ as

$$\begin{aligned} \xi_{Ak}(\tau) &= \sqrt{P_{Ak}}R(-\Delta\tau_{Ak} + \tau) \exp(j\Delta\theta_{Ak}) \\ \xi_{Ik}(\tau) &= \sqrt{\eta_k P_{Ak}}R(-\Delta\tau_{Ik} + \tau) \exp(j\Delta\theta_{Ik}) \end{aligned}$$

where P_{Ak} and η_k are the average values of P_A and η over the accumulation interval, and $\Delta\tau_{Ak}$ is the average value of the difference $\tau_A - \hat{\tau}$ over the accumulation interval, with similar definitions for $\Delta\tau_{Ik}$, $\Delta\theta_{Ak}$, and $\Delta\theta_{Ik}$.

The thermal noise component of the correlation function, $\xi_{Nk}(\tau)$, is modeled as having independent in-phase (real) and quadrature (imaginary) components, each modeled as a zero-mean Gaussian white discrete-time process:

$$E[\mathbb{R}\{\xi_{Nk}(\rho)\}\mathbb{I}\{\xi_{Nj}(\nu)\}] = 0 \quad \forall \rho, \nu, k \neq j$$

The chip interval of the spreading code $C_r(t)$, denoted τ_c , ranges from 0.01 to 1 μ s in modern GNSS signals. Due to the pseudorandom nature of $C_r(t)$, only samples of $\xi_{Nk}(\tau)$ within $2\tau_c$ of each other are correlated [33]:

$$E[\xi_{Nk}(\rho)\xi_{Nk}^*(\nu)] = \begin{cases} 2\sigma_N^2(1 - |\rho - \nu|/\tau_c) & |\rho - \nu| \leq 2\tau_c \\ 0 & |\rho - \nu| > 2\tau_c \end{cases}$$

Here, $*$ denotes the complex conjugate and σ_N^2 is the variance of the in-phase and quadrature components of $\xi_{Nk}(\tau)$, which is related to the spectral density of the white noise process $r_N(t)$ by $\sigma_N^2 = (N_0 + M_0)/2T$.

III. HYPOTHESIS TESTING FRAMEWORK

We adopt a Bayesian M-ary hypothesis testing framework [34, Ch. 2] for distinguishing between hypotheses H_i , $i \in \mathcal{I}$, where $\mathcal{I} = \{0, 1, 2, 3\}$. The null hypothesis H_0 corresponds to the interference-free case, and H_i , $i = 1, 2, 3$ correspond respectively to multipath, spoofing, and jamming.

The foregoing signal models reveal three parameters relevant to choosing between hypotheses, namely, the interference power advantage η , and the interference-to-authentic code and carrier offsets $\Delta\tau \triangleq \tau_I - \tau_A$ and $\Delta\theta \triangleq \theta_I - \theta_A$. Let these be combined into a single vector $\theta = [\eta, \Delta\tau, \Delta\theta]^T$ assumed to lie in the parameter space Λ , itself divided into disjoint parameter sets $\Lambda_i \subset \Lambda$, $i \in \mathcal{I}$, each associated with its corresponding

hypothesis H_i . Thus, deciding that $\theta \in \Lambda_i$ is equivalent to choosing hypothesis H_i . Note that, because θ can take on a range of values, the hypothesis testing problem is composite.

In a Bayesian formulation of the composite hypothesis testing problem, the parameter vector θ is viewed as a random quantity, Θ , having density $w(\theta)$, with $\pi_i \triangleq P(\Theta \in \Lambda_i)$ being the prior probability that Θ falls in Λ_i . We denote by $w_i(\theta)$ the conditional density of Θ given that $\Theta \in \Lambda_i$; it follows that

$$w_i(\theta) = \begin{cases} 0 & \theta \notin \Lambda_i \\ w(\theta)/\pi_i & \theta \in \Lambda_i \end{cases}$$

We propose to decide between the four hypotheses based on two types of observation at each t_k , namely, the received power measurement P_k and the symmetric difference measurement D_k , both detailed in a later section. The observation vector $z_k = [D_k, P_k]^T$, which resides in the observation set Γ , is modeled as a random variable Z_k with conditional density $p(z_k|\theta)$. H_i can be defined as the hypothesis that Z_k is distributed as $p(z_k|\Theta \in \Lambda_i)$, $i \in \mathcal{I}$.

A decision rule $\delta(z_k)$ is a partition of Γ into disjoint decision regions Γ_i , $i \in \mathcal{I}$, such that H_i is chosen when $z_k \in \Gamma_i$:

$$\delta(z_k) = \begin{cases} 0 & \text{if } z_k \in \Gamma_0 \\ 1 & \text{if } z_k \in \Gamma_1 \\ 2 & \text{if } z_k \in \Gamma_2 \\ 3 & \text{if } z_k \in \Gamma_3 \end{cases}$$

Let $C[i, \theta]$ be the cost of choosing H_i when $\theta \in \Lambda$ is the actual parameter vector. Note that this function is sensitive to a particular value of θ , which makes it more general than one that simply assigns a uniform cost for choosing H_i when $\theta \in \Lambda_j$. A later section will introduce various embodiments of $C[i, \theta]$.

An optimum rule selects the least costly hypothesis, on average, given the observation z_k . More precisely, if we define the conditional risk, or the average cost for $\Theta = \theta$, as

$$R_\theta(\delta) \triangleq E_\theta \{C[\delta(Z_k), \theta]\}, \quad \theta \in \Lambda$$

where E_θ denotes expectation assuming $Z_k \sim p(z_k|\theta)$, and if we define average, or Bayes, risk as

$$r(\delta) = E[R_\Theta(\delta)] \quad (6)$$

where the expectation is now taken over the random quantity Θ , then the optimum rule δ is the one whose decision regions Γ_i , $i \in \mathcal{I}$, minimize $r(\delta)$.

To find the minimizing δ , each parameter set Λ_i and conditional distribution $w_i(\theta)$ must be defined for $i \in \mathcal{I}$. These could be approximated from an extensive campaign of empirical multipath, spoofing, and jamming data collection and analysis. But any empirical characterization runs the risk of biasing the Λ_i and $w_i(\theta)$ toward the particular dataset used. This is merely a restatement of Hume's problem of induction: metaphorically speaking, the empirical dataset may only contain white swans [35]. To avoid this pitfall of inference, which, in a security context, could be a serious vulnerability, the following definitions of Λ_i and $w_i(\theta)$ are informed not only by observation but also by the physical characteristics and limitations of signals under the various hypotheses.

H_0 : No interference

In what follows, we denote the marginal distributions of $w_i(\theta)$ by $w_{\eta i}(x)$, $w_{\Delta \tau i}(x)$, and $w_{\Delta \theta i}(x)$, $i \in \mathcal{I}$. Under the null hypothesis ($i = 0$), which corresponds to the interference-free case, we have

$$\Lambda_0 = \{\theta \in \Lambda \mid \eta = 0\}$$

It follows that $w_{\eta 0}(x)$ is equivalent to the Dirac delta function. The marginal distributions $w_{\Delta \tau 0}(x)$, and $w_{\Delta \theta 0}(x)$ can be defined arbitrarily, since for $\eta = 0$ they have no effect on $r(\delta)$.

H_1 : Multipath

In the case of multipath, η and $\Delta \tau$ can be bounded as follows:

$$\Lambda_1 = \{\theta \in \Lambda \mid 0 < \eta < \eta_1, 0 < \Delta \tau < \Delta \tau_1\}$$

The bounds η_1 and $\Delta \tau_1$ are informed by the physics of signal reception and signal processing, as follows. Whenever the authentic signal is unobstructed, it arrives with greater power than any echo, whose additional path length and interaction with reflection surfaces invariably attenuate its power [36]. Moreover, an echo whose delay is more than double the spreading code chip interval (e.g., more than $0.2 \mu\text{s}$ for $\tau_c = 0.1 \mu\text{s}$) causes no correlation function distortion [33], and, due to the additional path loss, is at least 3 dB weaker than an unobstructed authentic signal [37]. Thus, its effect on $r(\delta)$ is negligible. It follows that $\eta_1 = 1$ and $\Delta \tau_1 = 2\tau_c$ upper-bound the multipath parameter set for an unobstructed authentic signal.

On the other hand, the severe shadowing experienced by mobile receivers can cause $\eta > 1$, especially in urban environments where surrounding buildings simultaneously attenuate and reflect authentic signals [37]. In these situations, there is no practical upper limit on η because the direct-path authentic signal may be attenuated by 50 dB or more. It is not possible to reliably distinguish multipath from low-power spoofing in such circumstances.

One can avoid this difficulty by applying this paper's detection test only when the authentic signal $r_A(t)$ is received without severe shadowing, in which case the probability that $\eta > 1$ under H_1 again becomes negligible. In particular, the multipath model developed below assumes that $r_A(t)$ is attenuated less than 6 dB by shadowing. In practice, one simply excludes cases in which the received power P_k is not unusually high yet the measured carrier-to-noise ratio C/N_0 drops by more than 6 dB from its modeled value for an unobstructed authentic signal.

An analysis of GNSS multipath was carried out to validate the bounds $\eta_1 = 1$ and $\Delta \tau_1 = 2\tau_c$ and to characterize $w_1(\theta)$. The analysis was based on the Land Mobile Satellite Channel Model (LMSCM) [38], itself based on extensive experimentation with a wideband airborne transmitter at GNSS frequencies in urban and suburban environments [37], [38]. The LMSCM generates power, delay, and carrier phase for both line-of-sight and echo signals using deterministic models for attenuation,

diffraction, and delay that respond to stochastically-generated obstacles and reflectors in the simulation environment.

In keeping with the philosophy of robust detection, multipath scenarios were sought whose distribution $w_1(\theta)$ was most similar to the distribution for spoofing, $w_2(\theta)$. From the standpoint of distinguishing multipath from spoofing, this is the worst-case $w_1(\theta)$. As a practical matter, the worst-case $w_1(\theta)$ has a high proportion of large η values and a wide range of $\Delta\theta$. Not surprisingly, urban LMSCM scenarios with low-elevation satellite signals yielded the worst-case $w_1(\theta)$.

For flexibility of detector design, the simulation-derived $w_1(\theta)$ was parameterized in terms of satellite elevation angle α_e . Several 1-minute LMSCM simulations with randomly-created urban environments and various satellite azimuth angles were run for $\alpha_e \in [20, 80]$ degrees. At each simulation epoch, the maximum-amplitude echo was identified and its relative power, delay, and phase with respect to the line-of-sight signal were taken as sampled η , $\Delta\tau$, and $\Delta\theta$ values from which $w_1(\theta)$ could be approximated. Epochs whose line-of-sight signal was attenuated by more than 6 dB were excluded, resulting in exclusion of up to 90% of epochs for heavily-shadowed scenarios at $\alpha_e = 20$, but less than 1% of epochs at $\alpha_e = 80$. At each α_e , the scenario producing the worst-case $w_1(\theta)$ was selected.

A statistical analysis of the sampled η , $\Delta\tau$, and $\Delta\theta$ values revealed that the relative phase $\Delta\theta$ was uniformly distributed on $[0, 2\pi)$ and independent of η , $\Delta\tau$, which is unsurprising given the wavelength-scale sensitivity of $\Delta\theta$ to path length. The parameters η and $\Delta\tau$ were found to be significantly correlated, with a linear correlation coefficient of approximately $\rho = -0.23$, implying that more distant reflectors tend to produce weaker echos. Consistent with [36], the marginal distribution $w_{\eta 1}(x)$ was found to be log-normal, with a mean of -21 dB and a standard deviation of 5 dB, both insensitive to α_e . Thus, excluding cases of heavy line-of-sight signal shadowing, multipath causes $\eta > 1$ with negligible probability, which indicates that $\eta_1 = 1$ is an unproblematic bound. The marginal distribution $w_{\Delta\tau 1}(x)$ was found to be exponential,

$$w_{\Delta\tau 1}(x) = \mu^{-1} e^{-x/\mu}, \quad x \geq 0$$

with μ a quadratic function of α_e ,

$$\mu = 0.012\alpha_e^2 - 2.4\alpha_e + 134$$

where α_e is expressed in degrees and $\Delta\tau$ in nanoseconds. Thus, $\Delta\tau$ values are widely spread at low α_e but tightly clustered near zero at high α_e , consistent with the empirical distributions in [37].

H₂: Spoofing

Under the spoofing hypothesis, we take $\eta_1 = 1$ as a lower bound for η and assume $|\Delta\tau|$ is upper-bounded by $\Delta\tau_1 = 2\tau_c$:

$$\Lambda_2 = \{\theta \in \Lambda \mid \eta_1 \leq \eta, 0 < |\Delta\tau| < \Delta\tau_1\}$$

These bounds make sense because the spoofing signal must be at least as powerful as the authentic signal for reliable spoofing, and because spoofing whose $|\Delta\tau|$ value exceeds $\Delta\tau_1 = 2\tau_c$ is no longer classified as carry-off-type spoofing,

since the interference is uncorrelated with the authentic signal. Instead, for $|\Delta\tau| \geq \Delta\tau_1$, the attack is classified as jamming.

For maximum stealth, the distribution $w_2(\theta)$ should be as close as possible to $w_1(\theta)$ while respecting the bounds η_1 and $\Delta\tau_1$ and allowing a wide range of $\Delta\tau$ for code pull-off. Thus, $w_{\eta 2}(x)$ is modeled as log-normal with mean of 1 dB and standard deviation a fraction of a dB, $w_{\Delta\tau 2}(x)$ as exponential with parameter $\mu = 120$ (mimicking low-elevation multipath), and $w_{\Delta\theta 2}(x)$ as uniform on $[0, 2\pi)$.

H₃: Jamming

Because it is uncorrelated with the authentic signal, jamming is taken to have $|\Delta\tau| \geq \Delta\tau_1 = 2\tau_c$. Its power advantage is assumed to be at least $\eta_1 = 1$, as weaker jamming is both harmless and so common as to be unremarkable. Thus, the jamming parameter set is

$$\Lambda_3 = \{\theta \in \Lambda \mid \eta_1 \leq \eta, \Delta\tau_1 \leq |\Delta\tau|\}$$

The marginal distribution $w_{\eta 3}(x)$ is taken to be Rician. By adjusting the Rician distance and scale parameters ν and σ , one can model high- or low-power jamming within a wide or narrow range. The marginal distributions $w_{\Delta\tau 3}(x)$ and $w_{\Delta\theta 3}(x)$ can be modeled arbitrarily, as they have no effect on $r(\delta)$ since $|\Delta\tau| \geq \Delta\tau_1$ eliminates correlation with the authentic signal.

IV. MEASUREMENT MODELS

This section develops models for the two observations that constitute the interference detection statistic, namely, the received power measurement and the symmetric difference measurement.

A. Received Power Measurement

The total received power measured by a GNSS receiver in an RF band of interest, denoted P_k for measurement time t_k , is a simple and effective indicator of interference [20], [39]. Even subtle spoofing attacks with power advantage η near unity can cause an increase in P_k that is distinguishable from random variations, including so-called nulling attacks that attempt to suppress the authentic signals [31]. The catch is that routine events such as solar radio bursts and the near passage of a so-called personal privacy device can also cause P_k to rise above nominal levels [31]. Thus, to prevent an anomalous-power monitor from issuing alarms intolerably often, the alarm threshold may need to be raised to a point where the monitor is no longer sensitive to subtle spoofing attacks. This is why power monitoring is best coupled with other forms of interference monitoring.

Suppose $M_s + 1$ authentic signals modeled as (1) are received, each with its associated interference signal, modeled as (2). Let $r_{Ai}(t)$ and $r_{Ii}(t)$ represent the i th authentic and interference signals. The combined signal exiting the RF front end is then

$$r_C(t) = \sum_{i=0}^{M_s} [r_{Ai}(t) + r_{Ii}(t)] + N(t) \quad (7)$$

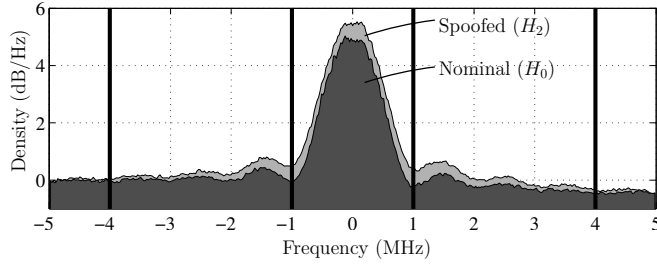


Fig. 3: Normalized power spectral density about the GPS L1 C/A center frequency (1575.42 MHz) for a static receiver platform during nominal conditions in a clean RF environment (dark gray), and during a spoofing attack (light gray). The two spectra are superimposed so that, of the spoofing attack spectrum, only the excess power beyond nominal is visible. The inner two vertical lines represent the 2-MHz bandwidth about L1, the outer two the 8-MHz bandwidth. The spoofing attack spectrum applies at $t_k = 130$ seconds into scenario 7 of the Texas Spoofing Test Battery (TEXBAT t_b7), described in [28].

where $N(t)$ is the same thermal noise component as in (3).

Receivers with sufficient dynamic range in the discrete samples produced from $r_C(t)$ do not require an AGC and so can compute received power directly by averaging the squared modulus of these samples. Let W_P be the bandwidth over which P_k is to be measured. If W_P is narrower than the receiver front-end's native bandwidth, then $r_C(t)$ must be filtered to isolate the desired spectral interval. Fig. 3 shows 2- and 8-MHz bands for P_k , with Fig. 4 showing corresponding power time histories. In a receiver whose front end is equipped with an AGC, P_k is measured indirectly through the AGC setpoint [20]. In this case W_P is equivalent to the front-end noise-equivalent bandwidth.

Let $\tilde{r}_C(t)$ represent the (optionally) filtered version of $r_C(t)$ from which P_k is to be measured. P_k is calculated as the average power in $\tilde{r}_C(t)$ over the interval t_{k-1} to t_k :

$$P_k \text{ (dBW)} \triangleq 10 \log_{10} \left(\frac{1}{T} \int_{t_{k-1}}^{t_k} |\tilde{r}_C(t)|^2 dt \right) \quad (8)$$

Let $p_{P_k}(x|\theta)$ be the conditional density of P_k . Assuming W_P is wide enough that $\tilde{r}_C(t)$ retains the significant power in $r_{Ai}(t)$ and $r_{Ii}(t)$, then $p_{P_k}(x|\theta)$ can be modeled as Gaussian, i.e., $p_{P_k}(x|\theta) = \mathcal{N}(x; \bar{P}_k, \sigma_P^2)$, where $\bar{P}_k$ and σ_P are expressed in dBW. The mean, $\bar{P}_k$, revealed by carrying out the multiplications inherent in $|\tilde{r}_C(t)|^2 = \tilde{r}_C(t)\tilde{r}_C^*(t)$, and by noting that $N(t)$ is zero-mean and independent of $r_{Ai}(t)$ and $r_{Ii}(t)$, and that, due to orthogonality of spreading codes, $E[r_{Ai}(t)r_{Ai}^*(t)] = E[r_{Ii}(t)r_{Ii}^*(t)] = 0$ for all $i \neq l$, is $\bar{P}_k = 10 \log_{10}(\bar{P}_L)$, where

$$\bar{P}_L = \sum_{i=0}^{M_s} [(1 + \eta_i)P_{Ai} + 2\sqrt{\eta_i}P_{Ai} \cos(\Delta\theta_i)R(\Delta\tau_i)] + N_0 W_P \quad (9)$$

and the subscript i associates the corresponding quantity with the i th signal. The first term in the summation is the non-

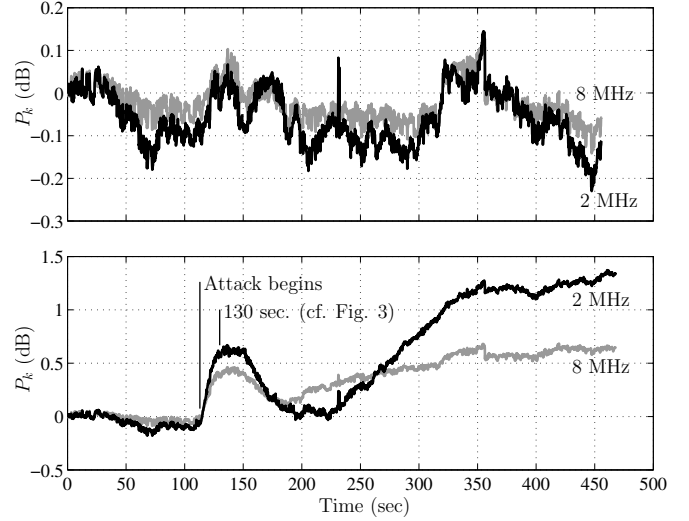


Fig. 4: Time histories of measured received power P_k for the nominal scenario (top) and spoofing scenario (bottom) whose spectra are shown in Fig. 3. Black traces correspond to the 2-MHz bandwidth and gray traces to the 8-MHz bandwidth centered at L1. The averaging interval for each measurement is 200 ms. P_k has been normalized to 0 dB at $t_k = 0$. The 130-second mark noted in the bottom panel is the instant at which the spoofing attack spectrum in Fig. 3 applies. At this point in the attack, $\eta_i \approx 4$, $\Delta\theta_i \approx \pi$, and $\Delta\tau_i \approx 0$ for $i = 0, \dots, M_s$.

coherent sum of power in r_{Ai} and r_{Ii} ; the second term is the power contributed by coherent interaction between r_{Ai} and r_{Ii} . If $\eta_i = 0$, or if the interference is orthogonal to the authentic signal (e.g., $\Delta\theta_i = \pm\pi/2$), or if $\Delta\tau_i$ is large enough that $R(\Delta\tau_i) = 0$, then this second term vanishes.

The deviation σ_P about $\bar{P}_k$ accounts for unpredictable but natural variations in P_k such as measurement error due to the finite duration over which P_k is estimated. From the top plot of Fig. 4, one can see that for the receiver under test, σ_P is at approximately 0.1 dB and 0.05 dB, respectively, for 2- and 8-MHz bandwidths. For urban RF environments, in which significant low-level interference is present in the GNSS bands, σ_P can be as high as 0.5 dB in a 2-MHz bandwidth. This paper's detection test assumes $\sigma_P = 0.4$ dB.

For modeling completeness, the multi-access interference $M(t)$ in (3) can be interpreted in terms of the model for $r_C(t)$ in (7). $M(t)$ models the effect of multi-access interference for a particular desired signal, $r_A(t)$. Invoking the so-called thermal-noise approximation [31], $M(t)$ is assumed to be spectrally flat with density M_0 , whose variable value is a function of the spreading code's chip interval, τ_c , and of the average power $\bar{P}_M$ and number M_s of multi-access signals. For GNSS signals with a $\text{sinc}^2(f\tau_c)$ -shaped power spectrum, which applies for all GPS signals except the new military M-code signals, M_0 is given by [31]

$$M_0 = (2/3)M_s\bar{P}_M\tau_c \quad (10)$$

Three example cases for $\bar{P}_M$ are given below. In each case, $i = 0$ indicates the desired authentic signal and its interferer, whereas $i = 1, \dots, M_s$ indicate multi-access signals and their

interferers.

$\bar{P}_M = P_A$: In this case, $P_{Ai} = P_A$ for $i = 0, \dots, M_s$ and $\eta_i = 0$ for $i = 1, \dots, M_s$. In other words, P_A is assumed to be typical of the received power from all GNSS signals and no other interference besides r_{I0} is assumed. This case applies for H_0 , for H_1 when only a single signal, r_{A0} , experiences significant multipath, and for H_2 when the spoofing attack targets only r_{A0} , as opposed to an attack against all r_{Ai} .

$\bar{P}_M = (1 + \eta)P_A$: In this case, $P_{Ai} = P_A$ and $\eta_i = \eta$ for $i = 0, \dots, M_s$. In other words, each of the $M_s + 1$ received signals is modeled as having power P_A , and each is assumed to be subject to a non-coherent interferer with power ηP_A . This case applies for the jamming hypothesis H_3 , as follows: For all i , $|\Delta\tau_i|$ is assumed to be large enough that $R(\Delta\tau_i) = 0$, causing the second term of the summation in (9) to vanish. Jamming power manifests in the first term of the summation as an additional $\eta_i P_{Ai} = \eta P_A$.

This case also applies for H_2 when the spoofing-to-authentic phase difference $\Delta\theta_i$ for $i = 0, \dots, M_s$ can be modeled as a random variable uniformly distributed on $[0, 2\pi)$, in which case the second term of the summation in (9) vanishes for all i .

$\bar{P}_M = (1 + \eta)P_A + 2\sqrt{\eta}P_A \cos(\Delta\theta)R(\Delta\tau)$: This case applies for H_2 when the spoofing attack targets not only r_{A0} but also the M_s multi-access signals such that $\eta_i P_{Ai} = \eta P_A$, $\Delta\theta_i = \Delta\theta$ and $\Delta\tau_i = \Delta\tau$ for $i = 0, \dots, M_s$.

B. Symmetric Difference Measurement

A variety of signal quality monitoring (SQM) metrics have been applied to detect distortions in $\xi_k(\tau)$ caused by anomalous GNSS signals, including spoofing signals [21], [22], [40]–[44]. Among these, the so-called symmetric difference is particularly attractive for multipath and spoofing detection because it is simple to implement, is insensitive to the particular shape of the correlation function (provided the function is symmetric about $\tau = 0$, as is the case for all current and proposed GNSS signals), and is sensitive to the correlation function distortion introduced by spoofing whenever the authentic and spoofing signals are approximately matched in power (i.e., $0.1 < \eta < 10$). Fig. 5 illustrates $\xi_k(\tau)$ under nominal conditions (top) and a spoofing attack (bottom).

Let τ_d be the offset of the early and late symmetric difference taps from $\tau = 0$, and let σ_{N0} be the value of σ_N in the interference-free case, for which we assume M_0 is given by (10) with $\bar{P}_M = P_A$. Then at measurement time t_k the symmetric difference is calculated as the modulus of the complex difference between the early and late symmetric difference taps, scaled by $1/\sigma_{N0}$:

$$D_k(\tau_d) \triangleq \frac{|\xi_k(-\tau_d) - \xi_k(\tau_d)|}{\sigma_{N0}} \quad (11)$$

As with P_k , a statistical model for $D_k(\tau_d)$ is necessary to develop a Bayesian detection framework. Let $p_{D_k}(x|\theta)$ denote the conditional density of D_k . An analytical model for $p_{D_k}(x|\theta)$ is easily derived when there is no coherent interaction between r_A and r_I , that is, when the second term of the summation in (9) vanishes, which occurs in the

interference-free case ($\eta = 0$) or in the case of jamming, for which $R(\Delta\tau) = 0$. Let P_{H_0} denote the received power when $\eta = 0$, which, as evident in the top panel of Fig. 4, is constant to within a few tenths of a dB. Assume that, for any $r(t)$ and any $k \in \{1, 2, \dots\}$, the AGC adjusts β_k such that the average power in $\beta_k r(t)$ over t_{k-1} to t_k is P_{H_0} . Then, if code tracking is accurate ($\Delta\tau_{Ak} \approx 0$), as would be true for up to moderate jamming, or if the amplitude of ξ_A is small with respect to σ_N , as would be true for strong jamming, then the scaled complex difference $[\xi_k(-\tau_d) - \xi_k(\tau_d)]/\sigma_{N0}$ can be modeled as a zero-mean complex Gaussian random variable with variance

$$\sigma_d^2 = 8\tau_d \left(\frac{\beta_k \sigma_N}{\sigma_{N0}} \right)^2 \quad (12)$$

It follows that $p_{D_k}(x|\theta)$ in this case is Rayleigh with scale parameter σ_d^2 :

$$p_{D_k}(x|\theta) = \frac{x}{\sigma_d^2} \exp\left(-\frac{x^2}{2\sigma_d^2}\right), \quad x \geq 0 \quad (13)$$

In the interference-free case, $\beta_k = 1$ and $\sigma_N = \sigma_{N0}$, so σ_d^2 reduces to $\sigma_d^2 = 8\tau_d$. One can show that this simplified expression for σ_d^2 is also approximately true for wideband jamming, for which the AGC maintains $(\beta_k \sigma_N / \sigma_{N0})^2 \approx 1$. However, the structured-signal jamming assumed in this paper is more potent than wideband jamming: it increases M_0 , and therefore σ_N , more than equivalently-powered wideband jamming [31]. As a result, σ_d^2 rises with increasing power of structured-signal jamming, making it more difficult to distinguish jamming from spoofing. Nevertheless, assuming structured-signal jamming, as opposed to the less potent wideband or narrowband jamming, is consonant with this paper's focus on worst-case attacks.

Note that a version of the symmetric difference normalized by the maximum magnitude of $\xi_k(\tau)$ has also been proposed for SQM and spoofing detection [23], [40], [41]. But the definition of D_k in (11) is more convenient because its distribution for $\eta = 0$ depends only on σ_d , as evident in (13).

Modeling $p_{D_k}(x|\theta)$ as Rayleigh only holds for $\eta = 0$ or $R(\Delta\tau) = 0$. When neither of these conditions is true, the real and imaginary components of $\xi_k(\tau)$ can manifest strong asymmetry, as shown in the bottom panel of Fig. 5. In this case, the possible interactions of $\xi_{Ik}(\tau)$, and $\xi_{Ak}(\tau)$ are so complex that $p_{D_k}(x|\theta)$ cannot be modeled analytically. It can, however, be studied through Monte-Carlo simulation of (5) and (11) as a function of $\theta = [\eta, \Delta\tau, \Delta\theta]^T$ and τ_d , together with some assumed recipe for determination of $\hat{\tau}$, the receiver's estimate of τ_A . For this paper, we have assumed that

$$\hat{\tau} = \arg \max_{\tau} |\xi_k(\tau)| \quad (14)$$

which holds for early-minus-late DLL tracking (both coherent and non-coherent) in the limit as $\tau_{DLL} \rightarrow 0$, provided the tracking points have not settled on a local maximum different from the global maximum. Note that to minimize the number of correlation taps, one can choose $\tau_{DLL} = \tau_d$, where τ_{DLL} is the offset from $\tau = 0$ of the DLL's early and late taps. But, depending on the shape of $\xi_k(\tau)$ and the assumed distribution

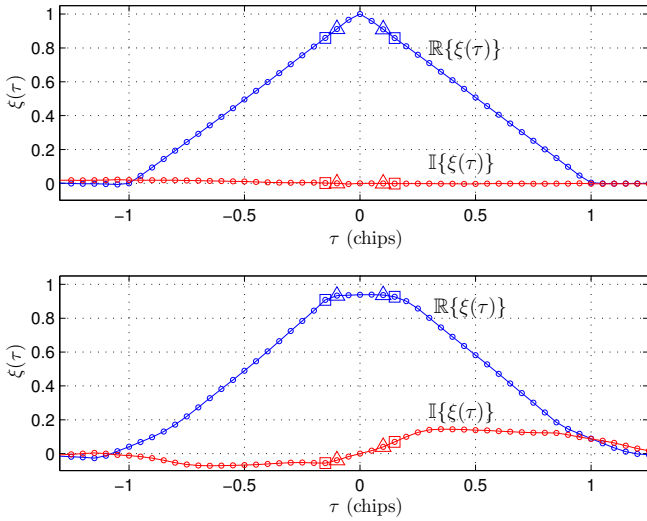


Fig. 5: In-phase $\mathbb{R}\{\xi_k(\tau)\}$ and quadrature $\mathbb{I}\{\xi_k(\tau)\}$ components of the empirical correlation function $\xi_k(\tau)$ for a clean scenario (top) and for a spoofing scenario (bottom). The early and late taps used for tracking [$\xi_k(-\tau_{\text{DLL}})$ and $\xi_k(\tau_{\text{DLL}})$; marked with $\square$] and for the symmetric difference measurement [$\xi_k(-\tau_d)$ and $\xi_k(\tau_d)$; marked with $\triangle$] are not necessarily the same. The nominal case (top) exhibits symmetry about $\tau = 0$ in both components, whereas the spoofed case (bottom) shows distortion and strong asymmetry in both components. Samples of the signal $r(t)$ have been scaled such that $|\xi_k(0)| = 1$ for the nominal case.

of τ_1 , unequal τ_{DLL} and τ_d may improve tracking and detection performance.

In simulation, one can apply whatever algorithm for producing $\hat{\tau}$ best approximates the operation of the receiver, or class of receivers, whose reaction to interference one wishes to simulate. For example, a multipath-mitigating receiver may estimate τ_A by (1) assuming the presence of one or more multipath components, (2) estimating the parameters η , τ_1 and θ_1 for each component, and (3) subtracting a model of each component from $\xi_k(\tau)$ [45].

C. Combined Measurement Vector

The symmetric difference for tracking channel $i \in \{1, 2, \dots, N\}$ at time t_k , denoted $D_k^i(\tau_d)$, can be combined with the received power measurement to form a channel-specific vector of observables at t_k :

$$z_k^i = [D_k^i(\tau_d), P_k]^T$$

$D_k^i(\tau_d)$ and P_k are modeled as independent so that

$$p(z_k^i | \theta) = p_{D_k^i}(z_{1k}^i | \theta) p_{P_k}(z_{2k}^i | \theta)$$

where z_{jk}^i is the j th element of z_k^i .

Obviously, when multiple GNSS signals are simultaneously affected by interference, detection performance improves by considering the multi-channel observation vector

$$z_k^{(1:N)} = [D_k^1(\tau_d), D_k^2(\tau_d), \dots, D_k^N(\tau_d), P_k]^T \quad (15)$$

For clarity of presentation and analysis, the channel-specific vector z_k^i has been taken as the observation vector for this paper's Bayesian detection strategy, with the superscript i dropped for notational simplicity. Extension of this paper's methods to the multi-channel case follows straightforwardly from the principles of the per-channel test along with a model for correlation between the $D_k^i(\tau_d)$.

V. THE POWER-DISTORTION TRADEOFF UNDER SPOOFING

The foregoing measurement models allow one to appreciate the power-distortion tradeoff that a spoofer faces when mounting an attack. In a typical attack, an admixture of spoofing and authentic signals is incident on the receiver's antenna. If the spoofing and authentic signals are approximately matched in power (i.e., $0.1 < \eta < 10$), then the correlation function will be significantly distorted, as shown in the lower panel of Fig. 5. The spoofer has several options for reducing this telltale distortion, but the alternatives are either intrinsically difficult, lead to anomalously high received power, or preclude effective capture, as described below.

A. Nulling or Blocking

A spoofer can minimize the hallmark distortions of an attack by generating an antipodal, or nulling, signal or by preventing reception of the authentic signal (e.g., by emplacing an obstruction). The form given for $r_1(t)$ in (2) accommodates a nulling attack in which $r_1(t)$ is made antipodal to $r_A(t)$ via the following settings: $\eta = 1$, $\tau_1 = \tau_A$, and $\theta_1 = \theta_A + \pi$. This attack annihilates $r_A(t)$, producing an effect similar to jamming but with much less received power. But the form in (2) does not accommodate the nulling-and-replacement attack described in [6] and [31], whereby, after nulling $r_A(t)$, the attacker supplants it with a separate spoofing signal. Detection of a perfectly-executed nulling-and-replacement attack is, in fact, not possible with the technique developed in this paper. Such an attack is, however, extremely difficult to carry out in practice, as it requires centimeter-level knowledge of, and a highly accurate fading model for, the attacker-to-receiver signal path.

Similarly, blocking reception of $r_A(t)$ by physical obstruction is difficult in cases where the receiver antenna is not physically accessible to the attacker. The technique developed in this paper makes the assumption that a nulling-and-replacement attack is impractically difficult and that physical access to the receiving antenna is controlled to prevent signal blockage. It must be recognized, however, that in many cases of practical interest (e.g., fishing vessel monitoring), the receiver and antenna are fully under the control of potential attackers.

B. Overpowering

An alternative approach to eliminating telltale distortion is an overpowered spoofing attack in which η is set so high that the receiver's AGC squelches the relatively weak authentic signals below the noise floor, thereby eliminating interaction between r_1 and r_A . In other words, as η increases, β_k decreases to maintain a constant power in $\beta_k r(t)$, with the result that

$\beta_k r_A(t)$ becomes negligible compared to $\beta_k r_I(t)$. If, however, the receiver raises an alarm when the received power P_k exceeds some threshold, then covert spoofing is strictly limited to $\eta < \eta_{\max}$ for some $\eta_{\max}$.

C. Underpowering

The spoofer can also minimize correlation function distortion by selecting a small η . However, reliable capture of the receiver's tracking loops requires $\eta > \eta_{\min} \approx 0.4$ dB [46]. Thus, for spoofing to be both reliable and covert, η is lower bounded by $\eta_{\min}$ and upper bounded by $\eta_{\max}$.

This paper's approach to spoofing detection can be stated as follows: If the defending receiver is designed such that $\eta_{\max}$ is sufficiently low despite maintaining a tolerable rate of false alarm in the received power monitor, then a carry-off spoofing attack that respects this bound yet successfully captures the receiver's tracking loops will unavoidably and detectably distort the correlation function $\xi_k(\tau)$. This distortion, evident in $D_k(\tau_d)$, can be distinguished from distortion due to multipath by its greater magnitude and by a concomitant increase in P_k .

VI. DECISION RULE

Finding the decision rule $\delta(z_k)$ from (6) that minimizes the Bayes risk $r(\delta)$ is conceptually straightforward. One can express $r(\delta)$ as

$$r(\delta) = E\{C[\delta(Z_k), \Theta]\} = E\{E\{C[\delta(Z_k), \Theta] | Z_k\}\} \quad (16)$$

where the second equality follows from the rule of iterated expectations. We note from (16) that, whatever the distribution of Z_k , $r(\delta)$ is minimized when, for each $z_k \in \Gamma$, δ is chosen to minimize the posterior cost $E\{C[\delta(Z_k), \Theta] | Z_k = z_k\}$. Thus, the Bayes-optimal rule is given by

$$\delta_B(z_k) = \arg \min_{i \in \mathcal{I}} E\{C[i, \Theta] | Z_k = z_k\} \quad (17)$$

which, because the parameter sets Λ_i are disjoint, can be written

$$\delta_B(z_k) = \arg \min_{i \in \mathcal{I}} \sum_{j=0}^3 \int_{\Lambda_j} C[i, \theta] p(\theta | z_k) d\theta \quad (18)$$

Reversing the conditioning of $p(\theta | z_k)$ using Bayes's formula, and recognizing that $w(\theta) = \sum_{j=0}^3 w_j(\theta) \pi_j$, we obtain

$$\delta_B(z_k) = \arg \min_{i \in \mathcal{I}} \sum_{j=0}^3 \pi_j \int_{\Lambda_j} C[i, \theta] p(z_k | \theta) w_j(\theta) d\theta \quad (19)$$

The foregoing sections defined the parameter sets Λ_i and presented analytical models for the conditional densities $w_i(\theta)$, $i \in \mathcal{I}$. If an analytical model for $p(z_k | \theta)$ were also available, then (19) could be solved by numerical integration. Unfortunately, finding an analytical model for $p(z_k | \theta)$ only appears possible for the special cases of $\eta = 0$ or $R(\Delta\tau) = 0$. On the other hand, for a given θ , it is easy to simulate samples drawn from $p(z_k | \theta)$ by taking the nonlinear and stochastic models described in Sections II and IV as recipes for a sample simulator. In view of this, a Monte-Carlo technique can be applied to find $\delta_B(z_k)$, as follows.

- 1) For each $i \in \mathcal{I}$, N_i parameter vectors are simulated according to the distribution $w_i(\theta)$, where $N_i/N_P \approx \pi_i$ and $N_P = \sum_{i \in \mathcal{I}} N_i$. The l th simulated vector drawn from $w_i(\theta)$ is denoted θ_{li} .
- 2) For each θ_{li} , N_M simulated measurements z_k are generated. Dropping the time index k from z_k for notational clarity, the m th simulated measurement, given θ_{li} , is written z_{mli} . The total number of measurements simulated is $N_P N_M$. Fig. 6 shows an example realization of this sample generation process over a relevant range for D_k and P_k .
- 3) The two-dimensional observation space Γ is divided into a large number of small rectangular cells of uniform size. Each cell is assumed to belong to a single decision region such that all observation samples falling within a cell belonging to Γ_i are assigned to hypothesis H_i , $i \in \mathcal{I}$. An initial partition of Γ is created by assigning each cell to the hypothesis H_i having the largest number of samples z_{mli} within the cell. Boundaries are adjusted such that each decision region Γ_i is simply connected (no islands), a condition that can reasonably be assumed from the fact that $C[i, \theta]$, $p(z_k | \theta)$, and $w_i(\theta)$ are smooth in θ .
- 4) The Bayes risk for the current partition is calculated as

$$r(\delta) = \frac{1}{N_P N_M} \sum_{i=0}^3 \sum_{l=1}^{N_i} \sum_{m=1}^{N_M} C[\delta(z_{mli}), \theta_{li}] \quad (20)$$

A new decision region assignment is considered for each cell lying along the boundary between decision regions. The new assignment is retained whenever it reduces $r(\delta)$, provided the resulting decision regions remain simply connected. The process is repeated until no boundary cells warrant re-assigning, at which point the cell assignments constitute the final decision regions Γ_i , $i \in \mathcal{I}$. Fig. 7 shows an example of the decision regions created by this process.

Two different types of cost function $C[i, \theta]$ are considered, as follows.

A. Uniform Cost within each Λ_j

When $C[i, \theta]$ is uniform across all $\theta \in \Lambda_j$, we write C_{ij} , interpreted as the cost of choosing H_i when H_j is true. The simplest such cost evenly penalizes misclassification: if $i = j$, then $C_{ij} = 0$; otherwise, $C_{ij} = 1$. But in the context of navigation security, not all types of misclassification are equally costly. We propose the following cost assignment:

- $C_{ii} = 0$, $i \in \mathcal{I}$: Correct decision
- $C_{01} = 0.2$: Low cost: Some multipath-induced code- and carrier- phase error might have been mitigated via receiver's multipath mitigation routines had multipath been detected, but otherwise harmless.
- $C_{02} = C_{12} = 1$: Highest cost: Spoofing goes undetected; receiver may report hazardously misleading information. Multipath mitigation applied by a receiver deciding H_1 cannot be assumed to reduce this cost.
- $C_{03} = C_{13} = 0.9$: High cost: Jamming goes undetected; receiver may report hazardously misleading information.

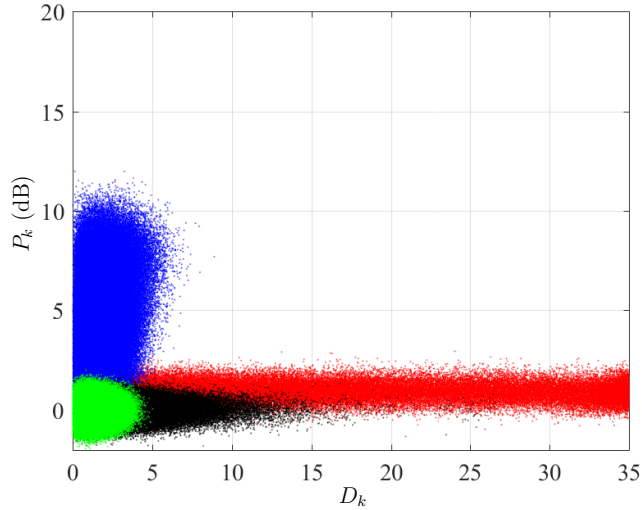


Fig. 6: Simulated observations $z_k = [D_k(\tau_d), P_k]^T$ for no interference (green), multipath (black), spoofing (red), and jamming (blue) based on the Monte-Carlo simulation technique described in the text, with $N_P = 10^5$, $N_M = 20$, $\pi_0 = 0.6$, $\pi_1 = 0.2$, $\pi_2 = 0.05$, $\pi_3 = 0.15$, $\sigma_P = 0.4$ dB, $P_A = -156$ dBW, $N_0 = -204$ dBW/Hz, $M_s = 7$, $\tau_d = 0.15$ chips, $T = 100$ ms, $W_P = 2$ MHz, and with $\bar{P}_M = P_A$ for H_0 and H_1 , $\bar{P}_M = (1 + \eta)P_A + 2\sqrt{\eta}P_A \cos(\Delta\theta)R(\Delta\tau)$ for H_2 , and $\bar{P}_M = (1 + \eta)P_A$ for H_3 . Note that the spoofing samples are concentrated along a low-power band because the distribution $w_2(\theta)$ assumes stealthy spoofing that attempts to masquerade as multipath.

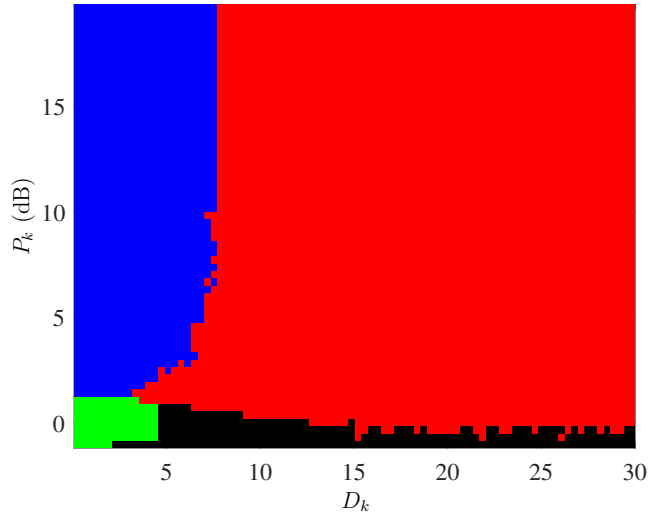


Fig. 7: Optimum decision regions for the θ -dependent cost $C[i, \theta]$: Γ_0 (no interference, green), Γ_1 (multipath, black), Γ_2 (spoofing, red), and Γ_3 (jamming, blue).

Not so costly as C_{02} because a jammer has less control over receiver output than a spoofer.

$C_{10} = 0.1$: Low cost: Receiver may waste computational resources trying to mitigate phantom multipath, and runs a chance of slightly biasing code and carrier measurements in the process, but otherwise harmless.

$C_{20} = C_{21} = C_{30} = C_{31} = 0.4$: Moderate cost: A false alarm for jamming or spoofing is raised, breaking navigation solution continuity.

$C_{32} = C_{23} = 0.2$: Low cost: Although spoofing is misclassified as jamming or vice-versa, no hazardous misleading information is issued, as the receiver suppresses its navigation solution when it decides H_2 or H_3 .

B. θ -Dependent Cost

The harm to a GNSS receiver and its dependent systems or users may not be uniform across all $\theta \in \Lambda_i$ for a given $\Lambda_i \subset \Lambda$, $i \in \mathcal{I}$. It is therefore worthwhile to consider the full θ -dependent cost $C[i, \theta]$. Our approach begins with the uniform cost assignment described above, substituting $C[i, \theta]$, $\theta \in \Lambda_j$ for certain elements C_{ij} , as follows:

$C[0, \theta \in \Lambda_1] = \min[0.8, e_\tau(\theta)/0.3]$: Not all multipath is equally harmful: Cost is assumed to increase linearly with the magnitude of multipath-induced code-phase error $e_\tau(\theta) \triangleq |\hat{\tau}(\theta) - \tau_A|$, expressed in chips of length τ_c , up to a saturation value of 0.8, where $\hat{\tau}(\theta)$ is given by (14) and $\min[x, y]$ is the minimum of x and y .

$C[0, \theta \in \Lambda_2] = C[0, \theta \in \Lambda_1]$ and $C[1, \theta \in \Lambda_2] = 0$ if $\Delta\tau < \tau_{\text{DLL}}$ and $\eta < -1$ dB; otherwise, $C[0, \theta \in \Lambda_2] = C_{02}$ and $C[1, \theta \in \Lambda_2] = C_{12}$, where τ_{DLL} is the offset of the correlation taps used for tracking (marked with $\square$ in Fig. 5): Close-in, weak spoofing is no more harmful than multipath, and can be treated accordingly.

$C[0, \theta \in \Lambda_3] = C[1, \theta \in \Lambda_3] = \min[C_{03}, \eta/10]$: The cost of jamming is assumed to increase linearly with η (in dB) up to saturation at C_{03} .

C. Prior Probabilities

The prior probabilities π_i , $i \in \mathcal{I}$, are of course situation-dependent. One may wish to increase π_2 , for example, upon entering an area where spoofer have historically been active. Values of π_i , $i \in \mathcal{I}$ were chosen as follows to represent a heightened threat scenario. Approximate relative values for π_0 , π_1 , and π_3 were first found by a manual epoch-by-epoch classification of data from a mobile GPS receiver in an urban environment, which exhibited significant multipath and mild jamming but no spoofing. A low but significant prior probability of spoofing was assumed, and the empirical π_1 and π_3 values were slightly inflated. All values were then normalized such that $\sum_{i \in \mathcal{I}} \pi_i = 1$. The resulting values, assumed for the remainder of the paper, were $\pi_0 = 0.6$, $\pi_1 = 0.2$, $\pi_2 = 0.05$ and $\pi_3 = 0.15$.

D. Application of the Decision Rule

The PD detector's Bayes-optimal decision rule δ_B for classifying received GNSS signals based on observations $z_k = [D_k(\tau_d), P_k]^T$, priors π_i , densities $w_i(\theta)$, $p(z_k|\theta)$, and the θ -dependent cost $C[i, \theta]$, all as described previously, and with parameter values as indicated in the caption of Fig. 6, is embodied in the colored regions shown in Fig. 7.

Application of the decision rule is straightforward: If the observation z_k^j , taken at time t_k from tracking channel j , falls

in decision region Γ_i , it is assigned to hypothesis H_i . Table I shows classification statistics as evaluated by applying the rule to an independent set of Monte-Carlo-generated observations like the one shown in Fig. 6. The table reveals that δ_B tends to misclassify multipath as H_0 , a consequence of the low cost $C[0, \theta \in \Lambda_1]$, especially when multipath is benign, as is often the case. Multipath is misclassified as spoofing less than 1.7% of the time even though the modeled $w_2(\theta)$ is unfavorable for distinguishing H_2 from H_1 .

One can achieve a lower spoofing false alarm rate by considering a set of channel-specific decisions over time $\{\delta_B(z_k^j) | k \in \mathcal{K}\}$ for some set $\mathcal{K}$ of contiguous sample times. This approach is effective because a spoofing attack must proceed slowly enough to capture the receiver tracking loops, which offers a window of several seconds over which $\{\delta_B(z_k^j) | k \in \mathcal{K}\}$ will include many declarations of H_2 , whereas, under H_1 , only a small number of such declarations would arise. One may also lower the spoofing false alarm rate by considering multi-channel decisions $\{\delta_B(z_k^j) | j = 1, 2, \dots, N\}$, or the full combined observation in (15). When applying multi-channel tests, one must bear in mind that a spoofer may not attack all channels simultaneously. However, any spoofer wishing to evade simple RAIM-type alarms must produce a self-consistent signal ensemble [47], which requires spoofing more than $N - N_m$ signals, where N is the total number of independent signals being tracked, and N_m (typically 4) is the minimum number required for a position and time solution.

TABLE I: Simulation-evaluated classification matrix for the decision regions in Fig. 7. The table's (i, j) th element is the relative frequency with which the detector chose i when j was the true scenario.

Decision	True Scenario			
	H_0	H_1	H_2	H_3
H_0	0.9942	0.8809	0.06244	0.0184
H_1	0.0005	0.0987	0.0234	0
H_2	0.0001	0.0162	0.8698	0.0017
H_3	0.0039	0.0028	0.0442	0.9799

VII. EXPERIMENTAL DATA

An independent evaluation of the PD detector was carried out against 27 empirical GNSS data recordings, including 6 recordings of various spoofing scenarios, 14 multipath-dense scenarios, 4 jamming scenarios of different power levels, and 3 scenarios exhibiting negligible interference beyond thermal and multi-access noise. Table II provides a summary description of each recording.

A. TEXBAT

The Texas Spoofing Test Battery (TEXBAT), version 1.1, from which *cd0* and *tb2*–*tb7* are drawn, is a public set of high-fidelity digital recordings of spoofing attacks against civil GPS L1 C/A signals [26]–[28]. Both static and dynamic scenarios are provided along with their corresponding unspoofed recording. Each 16-bit quantized recording is centered at 1575.42 MHz with a bandwidth of 20 MHz and a complex

sampling rate of 25 Msps. Each spoofing scenario makes use of the most advanced civil GPS spoofer publicly disclosed [3], [48]. In the laboratory, the spoofer can precisely control η , $\Delta\tau$, and $\Delta\theta$, and can generate self-consistent and aligned navigation data bits.

Note that *tb2*–*tb6* exhibit a modest amount of quantization and aliasing noise that makes them easier to detect than would be expected based on this paper's models, whereas *tb7* is free from such extraneous noise. *tb7* is also special in that the spoofer exercises control of $\Delta\theta$, permitting nulling, as described in Section V-A, in the early stages of the attack. In all other TEXBAT recordings, the spoofer controlled η and $\Delta\tau$ but left $\Delta\theta$ at an arbitrary constant value (*tb3*, *tb4*, and *tb6*) or allowed it to ramp consistent with the pull-off rate (*tb2* and *tb5*).

B. RNL Multipath and Interference Recordings

This public set of GNSS recordings, from which *wd0*–*wd12* and *sm1*–*sm3* are drawn, exhibits mild-to-severe multipath and mild unintentional jamming [29], [30]. Static and dynamic scenarios are included in both light and dense urban environments around Austin, TX. Each 16-bit quantized recording is centered at 1575.42 MHz with a bandwidth of 10 MHz and at a complex sampling rate of 12.5 Msps.

C. Intentional Jamming Recordings

Jamming scenarios *jd1*–*jd4* were recorded using a personal privacy device generating a sawtooth interference waveform with a sweep range of 1550.02–1606.72 MHz and sweep period of 26 μ s (see [49], Table 1, Row 1 and Fig. 8.). This device typifies low-cost jammers that can be purchased online and easily operated, albeit illegally. The jamming interference was combined with clean, static receiver data from a rooftop antenna and re-recorded. Each 16-bit quantized recording was centered at 1575.42 MHz with a bandwidth of 10 MHz and at a complex sampling rate of 12.5 Msps.

D. Clean Recordings

Three recordings (*cd0*, *wd0*, and *wd1*), with negligible interference beyond thermal noise and multi-access interference, were selected from the TEXBAT and RNL Multipath and Interference Recordings data sets. These were all static data sets in quiet RF environments with little multipath.

E. Pre-Processing

Raw wideband complex samples were first processed by the GRID science-grade software-defined receiver [50] to produce 100-Hz complex accumulations free of navigation data modulation at 41 uniformly-spaced taps spanning the range $[-\tau_c, \tau_c]$ around the prompt tap ($\tau = 0$). From the resulting data-free 100-Hz accumulations, the nominal thermal noise deviation σ_{N0} needed to form D_k in (11) was estimated by taking a complex-accumulation-wise standard deviation across multiple channels and multiple correlation tap offsets over a 15 second period in each recording that was substantially free from interference. Blocks of 10 100-Hz

TABLE II: Summary of 27 data recordings used for experimental evaluation.

Type	ID	Description	Duration (s)
H_0	cd0	static rooftop	456
H_0	wd0	static open field	304
H_0	wd1	static open field	298
H_1	wd2	dynamic deep urban	298
H_1	wd3	dynamic deep urban	456
H_1	wd4	dynamic deep urban	292
H_1	wd5	dynamic deep urban	292
H_1	wd6	dynamic deep urban	460
H_1	wd7	dynamic deep urban	246
H_1	wd8	dynamic deep urban	278
H_1	wd9	dynamic deep urban	334
H_1	wd10	dynamic deep urban	370
H_1	wd11	dynamic deep urban	390
H_1	wd12	dynamic deep urban	390
H_1	sm1	static urban	1700
H_1	sm2	static urban	600
H_1	sm3	static urban	150
H_2	tb2	static time push, $\eta = 10$ dB	346
H_2	tb3	static time push, $\eta = 1.3$ dB	336
H_2	tb4	static pos. push, $\eta = 0.4$ dB	336
H_2	tb5	dynamic pos. push, $\eta = 9.9$ dB	304
H_2	tb6	static time push, $\eta = 0.8$ dB	308
H_2	tb7	static time push, $\Delta\theta$ control	468
H_3	jd1	PPD, $\eta = 18$ dB	108
H_3	jd2	PPD, $\eta = 7$ dB	58
H_3	jd3	PPD, $\eta = 8$ dB	108
H_3	jd4	PPD, $\eta = 2$ dB	108

complex accumulations were then averaged to create 10-Hz accumulations, at each of the 41 code phase offsets. These 10-Hz complex accumulations are those modeled by $\xi_k(\tau)$ in Fig. 1.

Received power was calculated directly from the raw wide-band (e.g., 25 Msps) complex samples by filtering to a bandwidth W_P of 2 MHz, as in Fig. 3, then averaging the squared modulus of the filtered samples over 200 ms to produce a 5-Hz time history of received power. This was aligned in time with the $\xi_k(\tau)$ and interpolated to produce a corresponding 10-Hz P_k .

Naturally-occurring low-level (less than 2 dB) unintentional jamming was present in many of the type- H_1 (multipath) data recordings taken in urban settings. Fig. 8 shows an example P_k time history exhibiting such jamming as intermittent spikes. The short data intervals containing these spikes were excised from these recordings to ensure they exhibited only multipath and thermal noise. Otherwise, the PD detector would classify the type- H_1 recordings as a mixture of multipath and jamming, which, although true, would complicate an analysis of misclassification using the empirical data. As shown in the bottom panel of Fig. 8, the cumulative distribution of power for each data set was used to identify the distribution tail, at whose boundary the excision threshold was set.

VIII. EXPERIMENTAL RESULTS

This section documents the experimental performance assessment of the PD detector, when applying the decision regions shown in Fig. 7, against the 27 recordings listed in Table II. Because tb7 is a special case that violates the PD detector's assumption against nulling, it will be treated separately in the discussion below.

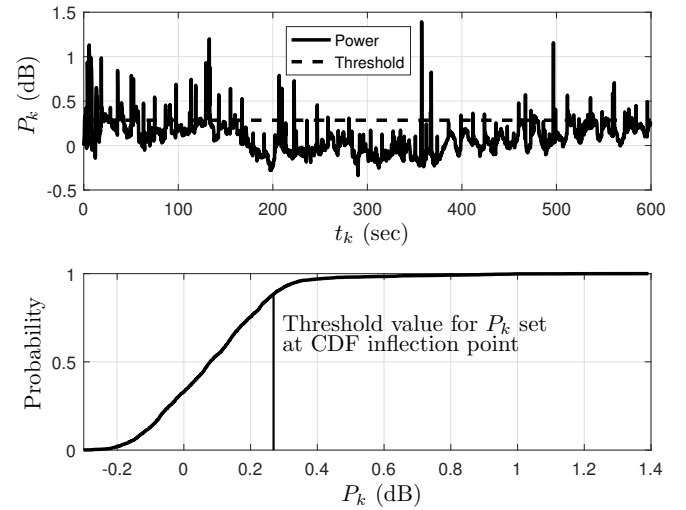


Fig. 8: Top: Time history of received power measurements P_k for data set sm2. Power spikes indicate the presence of low-level unintentional jamming. Bottom: The inflection point of the empirical cumulative distribution function (CDF) was taken as the threshold above which the empirical data were deemed to be of type H_3 (jamming) instead of type H_1 (multipath), and so excised from the recording.

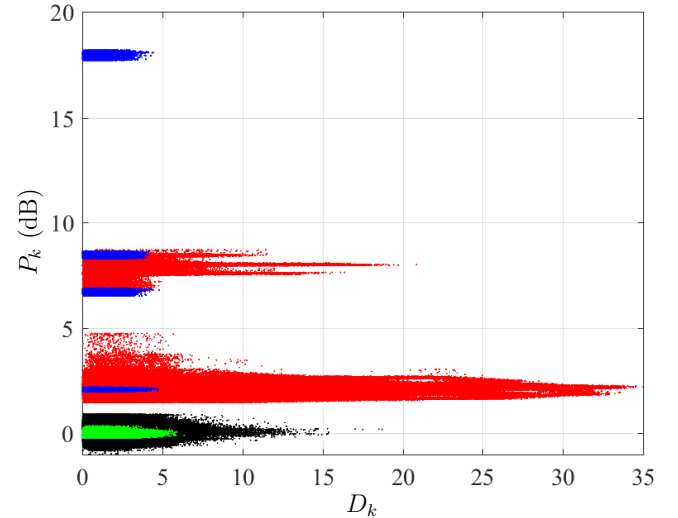


Fig. 9: Observations $z_k = [D_k, P_k]^T$ for clean (green), multipath (black), spoofing (red), and jamming (blue) from the experimental recordings.

Observations z_k from the experimental recordings (sans tb7) are shown in Fig. 9. Clearly, the simulated observations in Fig. 6 agree well with the empirical ones. The abrupt upper boundary of the multipath samples is due to the thresholding discussed in connection with Fig. 8.

Fig. 10 shows a single-channel cumulative time history of the PD detector's decisions for example jamming (top panel) and spoofing (bottom panel) attack scenarios. In the jamming scenario, the attack is detected immediately at onset, and continuously declared so thereafter. In the spoofing scenario, the attack is detected immediately, but initially classified as

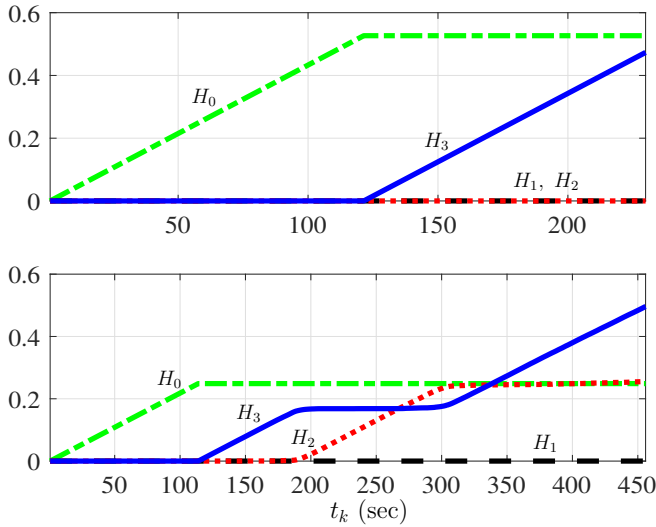


Fig. 10: Cumulative time history of decisions $\delta_B(z_k)$ for a single receiver tracking channel. Each trace represents the total number of times the corresponding hypothesis was chosen up to time t_k , normalized so that the final cumulative values sum to one. Both attacks begin at 120 seconds. Top: Jamming scenario j_d3 . Bottom: Spoofing scenario t_b4 with the detector applied to a mid-elevation satellite signal.

jamming because the spoofer's near-perfect initial code-phase alignment ($\Delta\tau \approx 0$) causes little distortion (and, indeed, little harm to the receiver). At about 180 seconds, the spoofer begins its pull-off, whereupon the increased correlation function distortion reveals the attack as spoofing. After $t_k = 300$, the pull-off has proceeded far enough that correlation function distortion is less pronounced and the attack is declared to be jamming again. It is notable that the spoofing attack is caught despite its low power advantage, which was $\eta = 0.4$ dB as intended, but ~ 1.5 dB in effect due to quantization noise in the spoofer.

Table III summarizes the PD detector's overall performance against the experimental data in terms of classification statistics. Importantly, all instances of spoofing and jamming were flagged as attacks. However, whereas jamming was always categorized correctly, spoofing was declared to be jamming 4/5 of the time, much higher than the simulated-data spoofing-to-jamming misclassification rate presented in Table I. The difference is that the experimental attacks all begin initially code-phase aligned ($\Delta\tau \approx 0$) and so do not initially cause harm or significant distortion. Moreover, as pull-off proceeds and $\Delta\theta$ exceeds τ_c , distortion subsides, becoming negligible beyond $2\tau_c$, whereupon the spoofing is either classified as H_0 (if low-power) or H_3 (if high-power). Thus, the PD detector is most effective at recognizing spoofing as such during initial carry-off of the tracking points. Conveniently, this is precisely when (1) carry-off-type spoofing begins to be hazardous, and when (2) civil GNSS spoofing detection strategies based on cryptographic security codes, as in [8], are least effective. Thus, the PD detector is nicely complementary with the methods of [8].

A second important result in Table III is the low rate of

false spoofing or jamming alarms. Clean (H_0) data produced no false alarms, and multipath-rich data (H_1) produced false alarms only 0.47% of the time. Thus, for a multi-channel decision, assuming an urban setting with independent multipath across tracking channels and $t_{k+1} - t_k = 100$ ms, any subset of 6 from a total of $N \leq 20$ tracking channels would only simultaneously false alarm on average once every 8.5 years.

TABLE III: As Table I but for the PD detector applied to the experimental recordings (*sans t_b7*).

Decision	True Scenario			
	H_0	H_1	H_2	H_3
H_0	0.9998	0.9253	0	0
H_1	0.0001	0.0700	0	0
H_2	0	0.0047	0.2013	0
H_3	0	0	0.7987	1

To test its limits, the PD detector was applied to t_b7 , an especially subtle attack in which the spoofer carefully controls $\Delta\theta$, effects authentic signal nulling during the initial stages of the attack, and maintains an approximately constant measured signal amplitude during pull-off [28]. As explained in Section V-A, such an attack would be difficult to mount outside the laboratory. As might be expected, the detector's performance was worse for this attack than for the other spoofing attacks: its decision rates during the attack portion of the recording were H_0 : 14%, H_1 : 10%, H_2 : 70%, and H_3 : 6%. Nonetheless, the attack was caught on each channel soon after pull-off began.

IX. CONCLUSIONS

We presented the PD detector, a novel low-cost, receiver-autonomous, readily-implementable GNSS jamming and carry-off spoofing detector. The detector traps a would-be attacker between simultaneous monitoring of received power and complex correlation function distortion. It amounts to a multi-hypothesis Bayesian classifier applied to a problem with three unknown parameters whose prior distributions are informed by the physics of GNSS signal reception and signal processing, and whose prior probabilities can be adjusted to reflect the threat environment in which a receiver operates. In evaluation against 27 high-quality experimental recordings of attack and non-attack scenarios, the detector correctly alarmed on all malicious attacks while maintaining a single-channel false alarm rate below 0.5%. For convenient implementation, the PD detector's decision rule for three different cost functions, together with all code required to generate application-tailored decision rules, is available at <https://github.com/navSecurity/P-D-defense>.

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